CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology (B.Tech) (EE)

University Examination November 2010

CE215 Object Oriented Programming

Date: 01.12.2010, Wednesday **Time: 10:30 a.m to 1:30 p.m** **Maximum Marks:70**

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1(a) Write the statements and mention whether the following are True or False. If false also provide the reason. **05**

1. Private members of the same class can directly access by the Class objects using the dot operator.
2. Member functions defined inside a class specifier become inline functions by default.
3. A class should have at least one constructor.
4. Qualification only reduces multiplicity from many to one.
5. A constructor is called whenever an object of the class is used.

Q.1(b) Answer the following questions. **06**

1. What is function overloading?
2. Explain Friend function with example.

Q.2 Answer the following questions. **12**

1. Write a program to add two complex numbers using operator overloading
2. Write a short note on Operators of C++.

OR

Q.2 Answer the following questions. **12**

1. Write a program to swap the private data of a class. Make a function exchange() for swapping. **06**
2. What is Inheritance? Explain different types of Inheritance. **04**
3. Explain *this* pointer. **02**

Q.3 Answer the following questions. **12**

1. Explain Inline Function with example.
2. Explain reference variable with example.
3. Explain copy constructor.

OR

Q.3 Answer the following questions. **12**

1. Explain New and Delete.
2. Explain structure of C++ Program.
3. Short Note: Benefits of OOP

SECTION – II

Q.4 Do as directed. **11**

1. Explain the following commands in MATLAB. **04**
   (a) whos (b) linspace (c) plot (d) clc
2. Write a computer graphics program to draw a circle in the center of the computer screen having radius 400 units. Also draw two lines passes through its center. Both lines should be at a 90 degree angle to each other. **04**
3. What is MATLAB? Draw its schematic diagram with main features. **03**

Q.5 Answer the following questions. **12**

1. Prepare an object diagram for a graphical document editor that supports grouping, which is a concept used in a variety of graphical editors. Assume that a document is composed of several sheets. Each sheet contains drawing objects, including text, geometrical objects, and groups. A group is simply a set of drawing objects, possibly including other groups. A group must contain at least two drawing objects. A drawing object can be a direct member of at most one group. Geometrical objects include circles, ellipses, rectangles, lines and squares.
2. Draw Event Trace Diagram for ATM scenario.
3. Differentiate:
   a) Query and Operation
   b) Association and Aggregation

OR

Q.5 Answer the following questions. **12**

1. Prepare a DFD to find volume and surface area of cylinder.
   Volume of cylinder = PI * r * r * h
   Surface Area of Cylinder = 2 * PI * r * h
2. What is scenario? What will be the scenario for two party phone call systems?
3. Differentiate:
   a) Event and State
   b) Activity and Operation

Q.6 Answer the following questions. **12**

1. Can Aggregation be fixed, variable or recursive? Why?
2. What is synergy? Also Define Role Name.
3. Explain: Module and Sheet
4. Define: Computer Graphics. Also explains its components.

OR

Q.6 Answer the following questions. **12**

1. What is the purpose of models in designing? What do you mean by views and stages of a model?
2. Group the following into association, aggregation and generalization:
   a) A diving philosopher is using a fork
   b) A polygon is order set of points
   c) A file is an ordinary file or a directory file.
3. What do you mean by Automatic transition? Also explain edge triggered event.
4. Is dynamic model a collection of state chart diagram? Why? Also separate out the following into active and passive components of DFD: processes, data store object, actor, data flows.

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